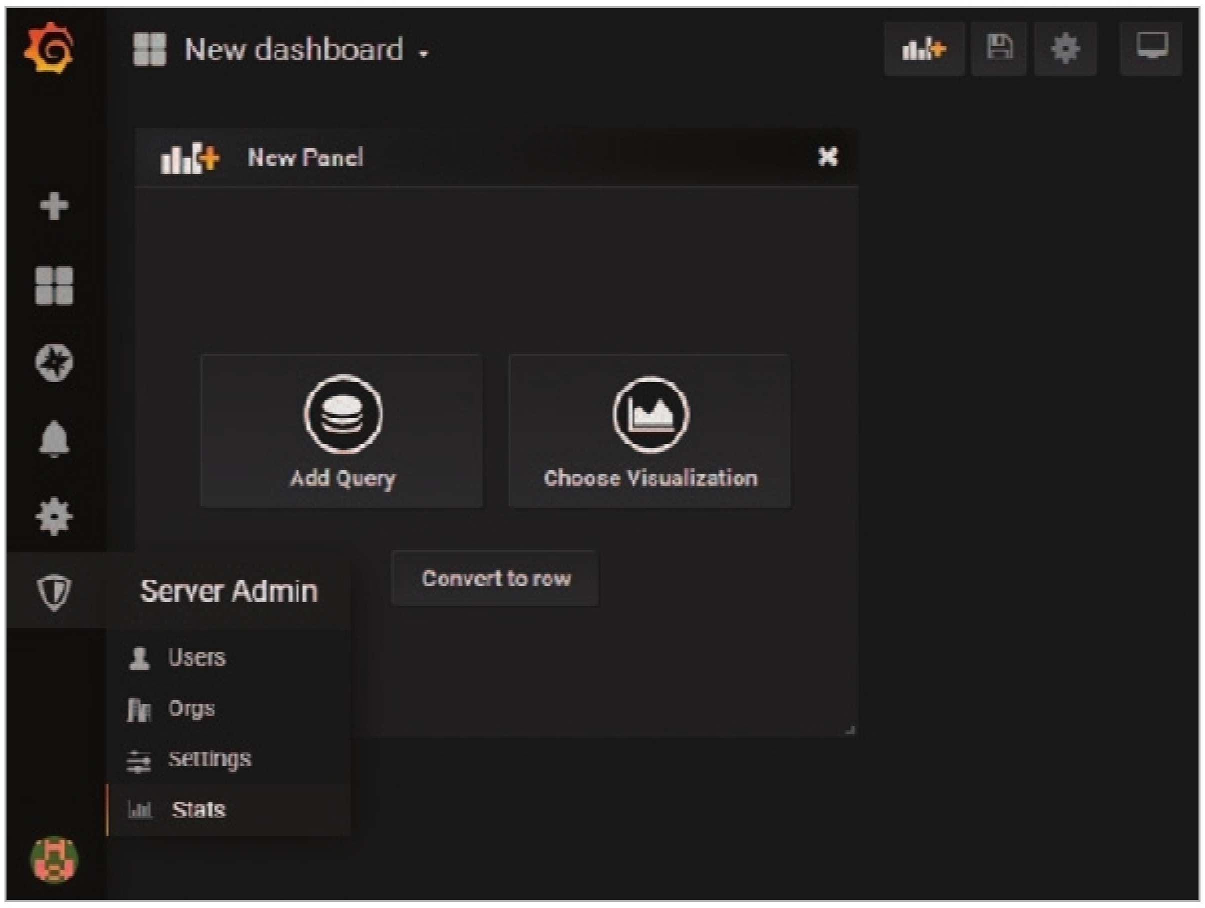

Figure 5: Data analytics and server administration in Grafana

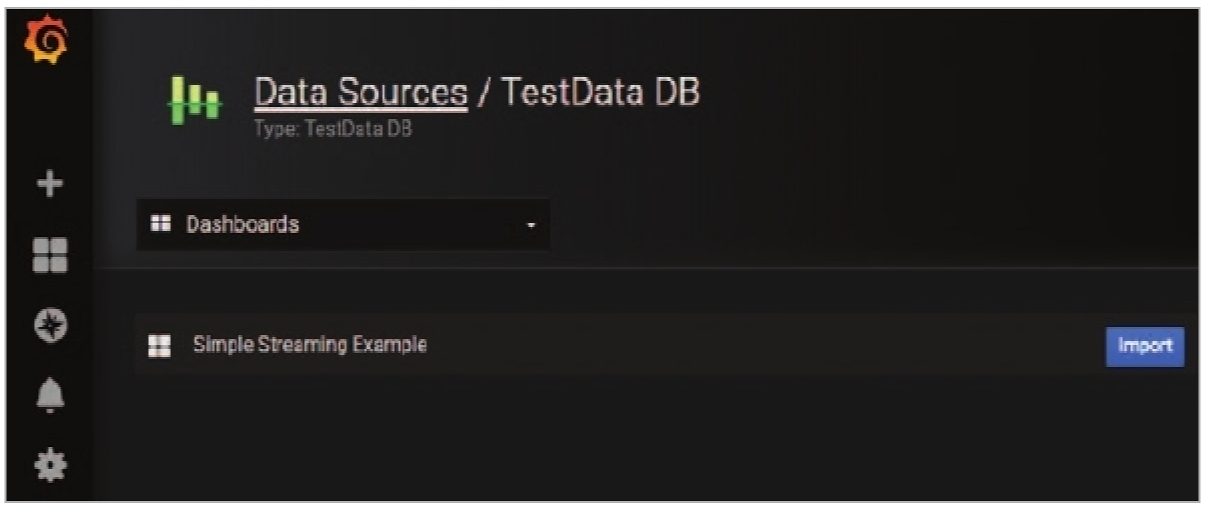

Figure 6: Built-in data channels and sources

Figure 7: Plotting and visualisation in Grafana

- Data dissemination, and sharing of dashboards and data

Grafana is being widely used for corporate applications in data analytics and the exploration of metrics using ad hoc queries on the data sources. There are many enterprise level implementations in various domains.

After downloading the Grafana library, the directory structure has a *bin* folder in which the *grafana-server* file is executed. On running the *grafana server* script, the Web browser based engine is activated and can be executed on any Web browser including Google Chrome, Mozilla Firefox, and Opera.

The Grafana Web browser based application uses the URL *http://localhost:3000* so that data science implementations can be done with the inbuilt visualisation and plotting, with cavernous analytics.

Grafana provides cloud compatibility so data scientists and researchers can directly use Grafana Cloud without manual installation. The cloud flavour of Grafana integrates the dashboard for data analytics and visualisation, in which different data sets can be analysed. The cloud version of Grafana is available free.

Grafana provides excellent power and features to import data sources from a wide range of database engines including Structured Query Language (SQL), Not Only SQL (NoSQL), JavaScript Object Notation (JSON), and many others.

The server administration panel in Grafana enables the evaluation of data with multiple and multi-dimensional statistics. The user management and logs analysis can be done in this panel, for security and integrity (Figure 4).

A number of additional data sources and plugins are distributed on *https://grafana.com/grafana/plugins*. These plugins can be integrated with the existing installation of Grafana for data analytics (Figure 5).

Research tasks in Grafana can be easily implemented using built-in templates, which can be used for data